

TAYLOR OCKERMAN

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PROFILE

BCIT Bachelor of Applied Computer Science student with experience developing gameplay systems, custom engine architecture, and real-time interactive applications across academic and personal projects. Strong background in C++, Unreal Engine, Unity, and collaborative game development workflows, with experience implementing gameplay features, UI systems, and performance-focused simulation systems. Interested in gameplay programming, technical design, and multidisciplinary game development.

RELEVANT TECHNICAL SKILLS

Game Development: Unreal Engine 5, Unity, C++, C#, Blueprints, SDL3, Gameplay Systems, UI Implementation, ECS Architecture, Git

Programming Languages: C/C++, C#, JavaScript, Python, TypeScript, Java

Tools & Frameworks: Git, React, Node.js, Express, HTML/CSS, MySQL

Technical Areas: Gameplay Programming, Real-Time Systems, Performance Optimization, Debugging, ObjectOriented Programming, Data Structures & Algorithms

PERSONAL PROJECTS

QuickStyle: In-browser React-based Tailwind CSS styling GUI

Winter 2026

- Placed first overall as a team of four in Hack the Break 2026 hackathon competition
- Identified limitations in existing Tailwind workflows and proposed a real-time visual editing tool
- Designed and built a React-based GUI to modify Tailwind styles dynamically
- Led implementation of core UI interaction features during a 40-hour hackathon

Gameplay Prototypes: Unreal Engine 5

Spring 2026

- Developed gameplay prototypes in Unreal Engine 5 using Blueprints
- Implemented projectile spawning systems, collision handling, and movement behaviors
- Built reusable gameplay actors with customizable firing and movement patterns
- Configured collision channels and gameplay interactions for player and environment systems

ACADEMIC PROJECTS

Sandy Sim: Custom C++ Game Engine (BCIT)

Winter 2026

- Designed and implemented a hybrid ECS/OOP architecture with modular systems for physics, rendering, input, and particle interactions in a cellular automata simulation
- Developed particle simulation systems supporting multiple material behaviors (liquids, gases, solids) with density and flammability-based interactions
- Optimized simulation performance using spatial partitioning (chunking), texture caching, and grid-based collision resolution
- Implemented layered rendering, dynamic UI systems, and camera-based scrolling using SDL3

Allez: Game Development (BCIT)

Fall 2025

- Headed the design and development of player combat systems, stamina mechanics, and gameplay logic using Unity and C#
- Implemented UI systems and gameplay feedback elements to support player experience and readability
- Collaborated within a five-person team using Git workflows and iterative development practices
- Participated in gameplay testing, balancing, debugging, and feature iteration throughout development

WORK EXPERIENCE

Laboratory Assistant – PCIGR (UBC)

May 2015 – Dec 2025

- Progressively entrusted with increased responsibility, including data processing, client reporting, and workflow optimization initiatives
 - Recognized for taking initiative and ensuring continuity of a critical client project under challenging circumstances
 - Defined and implemented Python scripts for mass spectrometer data import and export
 - Spearheaded workflow optimizations that led to a 5x increase in mineral imaging speed
 - Processed and validated large scientific datasets for accuracy and consistency
 - Collaborated with research staff to troubleshoot and resolve data quality concerns
 - Organized and tracked the status of hundreds of client samples throughout processing workflows
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EDUCATION

British Columbia Institute of Technology

Bachelor of Applied Computer Science

Sep 2025 – present

- CGPA: 94%

Computer Systems Technology Diploma

Graduated: Mar 2025

- CGPA: 93%